

Chapter 3—Extensions to Mendel: Complexities in Relating Genotype to Phenotype

Fill in the Blank

1. The proportion of all copies of a gene in a population that are of a given allele type is called _____.
Ans: allele frequency
Difficulty: 1
2. A wild-type allele is an allele designated by a superscript plus sign (+) whose frequency is greater than _____ percent.
Ans: one
Difficulty: 1
3. A gene with only one wild-type allele is called _____.
Ans: monomorphic
Difficulty: 2
4. A _____ is a non-inherited change in phenotype arising from environmental affects that mimic the effect of a mutation in a gene.
Ans: phenocopy
Difficulty: 2
5. _____ refers to traits determined by two or more factors, including multiple genes interacting with each other or one or more genes interacting with the environment.
Ans: multifactorial
Difficulty: 1
6. _____ genes produce a subtle, secondary effect on phenotype by altering the effect of the alleles of other genes.
Ans: modifier
Difficulty: 1
7. A _____ trait or continuous trait is an inherited trait that exhibits many intermediate forms; determined by segregating alleles of many different genes whose interaction with each other and the environment produces the phenotype.
Ans: quantitative
Difficulty: 2
8. A phenomenon in which a single gene determines a number of distinct and seemingly unrelated characteristics is called _____.
Ans: pleiotropy
Difficulty: 1

9. _____ is the degree or intensity with which a particular genotype is expressed in a phenotype.

Ans: expressivity

Difficulty: 1

10. If all individuals carrying a dominant mutant gene show the mutant phenotype, the gene is said to show complete _____.

Ans: penetrance

Difficulty: 1

Multiple Choice

11. The interaction between non-allelic genes that results in the expression of a phenotype is:

- A) epistasis.
- B) epigenetics.
- C) dominance.
- D) codominance.
- E) incomplete dominance.

Ans: A

Difficulty: 1

12. Which of the following diseases show pleiotropism?

- A) albinism
- B) muscular dystrophy
- C) colorblindness
- D) sickle cell anemia
- E) male pattern baldness

Ans: D

Difficulty: 1

13. A deviation from normal Mendelian ratios which may be resolved by counting and/or controlled crosses, is seen in which of the following terms?

- A) pleiotropy
- B) codominance
- C) incomplete dominance
- D) complete dominance
- E) penetrance and expressivity

Ans: E

Difficulty: 1

14. Which of the following ratios show incomplete dominance?

- A) 2:1
- B) 3:1
- C) 1:2:1
- D) 1:1
- E) 4:1

Ans: C

Difficulty: 2

15. Which of the following ratios show codominance?

- A) 2:1
- B) 3:1
- C) 1:2:1
- D) 1:1
- E) 4:1

Ans: C

Difficulty: 1

16. Which of the following ratios indicates a lethal gene?

- A) 2:1
- B) 3:1
- C) 1:2:1
- D) 1:1
- E) 4:1

Ans: A

Difficulty: 2

17. A person who has type O blood has:

- A) anti-A antibodies.
- B) anti-B antibodies.
- C) anti-AB antibodies.
- D) both anti-A and -B antibodies.
- E) no surface antigens.

Ans: D

Difficulty: 2

18. If two or more forms of the same gene exist, the different forms are called:

- A) incomplete dominance.
- B) penetrance and expressivity.
- C) pleiotropic.
- D) multiple alleles.
- E) dihybrid.

Ans: D

Difficulty: 2

19. The blood groups A, B and O are:

- A) incomplete dominance.
- B) penetrance and expressivity.
- C) pleiotrophic.
- D) multiple alleles.
- E) dihybrid.

Ans: D

Difficulty: 1

20. The blood groups A, B and O show:

- A) complete dominance.
- B) recessive.
- C) codominance.
- D) a, b, and c
- E) none of the above

Ans: D

Difficulty: 2

21. Which of the following monohybrid ratios can describe incomplete and codominance?

- A) 2:1
- B) 3:1
- C) 1:3
- D) 1:2:1
- E) 4:1

Ans: D

Difficulty: 1

22. Which of the following ratios demonstrate gene interaction?

- A) 2:1
- B) 3:1
- C) 1:2:1
- D) 9:3:4
- E) 1:3

Ans: D

Difficulty: 1

23. A _____ results whenever the nucleotide sequence is changed.

- A) phenotype
- B) genotype
- C) mutation
- D) trait
- E) character

Ans: C

Difficulty: 1

24. When the same gene is related to respiratory problems and sterility, it can be described as:

- A) pleiotropy.
- B) codominance.
- C) incomplete dominance.
- D) complete dominance.
- E) penetrance and expressivity.

Ans: A

Difficulty: 1

25. Another name for a normal gene is:

- A) wild-type.
- B) pleiotropy.
- C) dominant.
- D) codominant.
- E) recessive.

Ans: A

Difficulty: 1

26. The genetic ratio 1:2:1 may indicate:

- A) complete dominance.
- B) codominance.
- C) epistasis.
- D) recessive lethal.
- E) b and c only

Ans: B

Difficulty: 1

27. The genetic ratio 3 : 1 may indicate:

- A) complete dominance.
- B) codominance.
- C) epistasis.
- D) incomplete dominance.
- E) b and c only

Ans: A

Difficulty: 1

28. The genetic ratio 2:1 may indicate:

- A) complete dominance.
- B) codominance.
- C) epistasis.
- D) recessive lethal.
- E) b and c only

Ans: D

Difficulty: 1

29. The genetic ratio 9:7 may indicate:

- A) complete dominance.
- B) codominance.
- C) epistasis.
- D) recessive lethal.
- E) b and c only

Ans: C

Difficulty: 1

30. The genetic ratio 9:3:4 may indicate:

- A) complete dominance.
- B) codominance.
- C) epistasis.
- D) recessive lethal.
- E) b and c only

Ans: C

Difficulty: 1

31. Which of the following show independent assortment?

- A) 9:3:3:1
- B) 9:7
- C) 9:3:4
- D) 13:3
- E) all of the above

Ans: E

Difficulty: 1

Matching

Write the correct phase of epistasis, pleiotropy, complete dominance, recessive, incomplete dominance, or co-dominance for each question.

32. equal expression

Ans: co-dominant

Difficulty: 2

33. partial expression

Ans: incomplete dominance

Difficulty: 2

34. expressed in the heterozygous or homozygous conditions

Ans: complete dominance

Difficulty: 2

35. blood type “AB”
Ans: co-dominant
Difficulty: 2
36. neither gene is fully expressed
Ans: incomplete dominance
Difficulty: 2
37. only expressed in the homozygous condition
Ans: recessive
Difficulty: 2
38. whenever the allele is present it is expressed
Ans: dominant
Difficulty: 2
39. gene interaction
Ans: epistasis
Difficulty: 2
40. a gene that has more than one effect
Ans: pleiotrophic
Difficulty: 2
41. blood type $I^A I^O$
Ans: co-dominant
Difficulty: 1

True or False

42. Multifactorial inheritance is when a phenotype arises as a result of the interaction of one or more genes with the environment and/or each other.
Ans: True
Difficulty: 1
43. The flower colors white, pink and red indicate codominant inheritance.
Ans: False
Difficulty: 2
44. When a late blooming pea and an early blooming pea are crossed and an intermediate phenotype occurs, this result would suggest incomplete dominant inheritance.
Ans: False
Difficulty: 2

45. In codominance, F₁ hybrids show the traits of both parents.
Ans: True
Difficulty: 1
46. Different alleles indicate unique genes.
Ans: False
Difficulty: 1
47. Mutations are the source of new alleles.
Ans: True
Difficulty: 1
48. A wild-type allele is any allele whose frequency is closest to 100%.
Ans: False
Difficulty: 2
49. A mutant allele is one that with a frequency less than 1%.
Ans: True
Difficulty: 2
50. Genes with more than one wild-type allele are termed polymorphic.
Ans: True
Difficulty: 2
51. The mouse agouti gene has one wild-type allele and several mutant alleles.
Ans: True
Difficulty: 2
52. The phenomenon of a single gene determining a number of distinct and seemingly unrelated characteristics is known as pleiotropy.
Ans: True
Difficulty: 1

Short Answer

53. In corn three dominant genes are necessary for aleurone color. The genotype B__D__R is colored. Any homozygous recessive for one gene is colorless. Predict the genotypes and phenotypes of the offspring of the cross BbDdRr × BbDdRr

Ans: Phenotype: 27 colored; 37 colorless

Ratio of Genotype

1	BBDDrr
2	BbDDrr
2	BBDdrr
4	BbDdrr
1	BBDDRr
2	BbDDRr
1	BBDDrr
2	BbDDrr
1	BBDDRr
2	BbDDRr
4	BbDdRr
4	BbDdRr
8	BbDdRr
2	BBddRr
4	BbddRr
2	bbDDRr
4	bbDdRr
2	bbddRr
1	BBDDRR
2	BbDDRR
2	BBdRRR
4	BbDdRR
1	bbDDRR
2	bbDdRR
1	BBddRR
2	bbDdRR
1	bbddRR

Difficulty: 4

54. In corn three dominant genes are necessary for aleurone color. The genotype B__D__R is colored. Any homozygous recessive for one gene is colorless. Predict the genotypes and phenotypes of the offspring of the cross BbDdRR × BbDdRR

Ans: Phenotype: 9 Color; 7 Colorless

Ratio of Genotype

1	BBDDRR
2	BbDDRR
2	BBdRRR
4	BbDdRR
1	bbDDRR
2	bbDdRR
1	BBddRR
2	bbDdRR
1	bbddRR

Difficulty: 3

55. In corn three dominant genes are necessary for aleurone color. The genotype B__D__R is colored. Any homozygous recessive for one gene is colorless. Predict the genotypes and phenotypes of the offspring of the cross BbDdRR × BbDdrr

Ans: Phenotype: 9 Color; 7 Colorless

Ratio of Genotypes

1	BBDDRr
2	BBdDrr
2	BbDDRr
4	BbDdRr
1	BBddRr
2	BbddRr
1	bbDDRr
2	bbDdRr
1	bbddRr

Difficulty: 4

56. In the rat the gene for the pigment (P) is dominant to no pigment (p). The gene for black (B) is dominant to the gene for cream (b). If a pigment gene (P) is absent, genes B and b are inoperative. Predict the genotypes and phenotypes of the F1 of a cross between a homozygous black rat and an albino homozygous for cream.

Ans:

<u>Genotype</u>	<u>Phenotype</u>
PpBb	Black

Difficulty: 1

57. In the rat the gene for the pigment (P) is dominant to no pigment (p). The gene for black (B) is dominant to the gene for cream (b). If a pigment gene (P) is absent, genes B and b are inoperative. Predict the genotypes and phenotypes of the F₂ of a cross between a homozygous black rat and an albino homozygous for cream.

Ans: 9 Black; 3 cream; 4 colorless

	<u>Genotype</u>	<u>Phenotype</u>
1	PPBB	Black
2	PPBb	Black
2	PpBB	Black
4	PpBb	Black
1	ppBB	colorless
2	ppBb	colorless
1	PPbb	cream
2	Ppbb	cream
1	ppbb	colorless

Difficulty: 2

58. In the common daisy the genes A and a, B and b represent two pairs of alleles acting on flower color. A and B are required for color. The alleles of these two genes show recessive epistasis. The two gene pairs together, thus show duplicate recessive epistasis. Predict the genotypes and phenotypes of the F₁ of a cross between two colorless plants, one homozygous for A and the other homozygous for B.

Ans: Genotype Phenotype
 AaBb Color

Difficulty: 1

59. In the common daisy the genes A and a, B and b represent two pairs of alleles acting on flower color. A and B are required for color. The alleles of these two genes show recessive epistasis. The two gene pairs together, thus show duplicate recessive epistasis. Predict the genotypes and phenotypes of the F₂ of a cross between two colorless plants, one homozygous for A and the other homozygous for B.

Ans: 9 Black; 7 colorless

	<u>Genotype</u>	<u>Phenotype</u>
1	AABB	Color
2	AABb	Color
2	AaBB	Color
4	AaBb	Color
1	aaBB	colorless
2	aaBb	colorless
1	AAbb	colorless
2	Aabb	colorless
1	aabb	colorless

Difficulty: 2

Corn in Alabama and Tennessee were both found to produce colorless aleurone in corn seeds. When these two plants were crossed the resulting F₁ had Purple Aleurone. When the F₁ was crossed to produce and F₂ the results were 270 purple and 210 Colorless aleurone.

60. What is the genotype of the F₁ progeny?

Ans: The F₁ is a heterozygote; BbDd from parents BBdd and bbDD

Difficulty: 1

61. What are the genotypes and corresponding phenotype of the F₂?

Ans:	<u>Genotype</u>	<u>Phenotype</u>
1	BBDD	Color
2	BbDD	Color
2	BBdd	Color
4	BbDd	Color
1	bbDD	colorless
2	bbDd	colorless
1	BBdd	colorless
2	Bbdd	colorless
1	bbdd	colorless

Difficulty: 3

62. Give the possible biosynthetic pathway for gene activity.

Ans: A - B -
Substrate → Intermediate → Color

aa B -
Substrate → xxxxxxxxx → Colorless

A - bb
Substrate → Intermediate → Colorless

Difficulty: 3

63. In poultry, if a Black Longshank male with feathered shanks is crossed with a Buff Rock female with unfeathered shanks the F₁ are all feathered and the F₂ show 90 feathered to 6 unfeathered. Infer the genotypes of the parents.

Ans: AABB × aabb; The ratio is a 15:1 which is a dihybrid ratio; therefore the parents are homozygous and produce a heterozygous F₁.

Difficulty: 2

The following five mothers, (a) through (e), with given phenotypes, each produced one child whose phenotype is described as to blood group (A,B,O), M or N antigens, and Rh factor. For each child, select as the father one of the five males whose genotypes are given.

(ii = Type O blood) [rr = rh⁻ & R = rh⁺]

	Maternal Phenotype	Child Phenotype	Genotype of Male
(a)	A M R	O M R	1. $I^A i$ MN rr
(b)	B N r	O N r	2. $I^B i$ MN RR
(c)	O M r	A MN R	3. iiNNrr
(d)	A N R	AB MN R	4. iiMMrr
(e)	AB MN r	A MN r	5. $I^A I^A$ MNRR

64. Child of Mother (a) = Father _____

Ans: 4

Difficulty: 3

65. Child of Mother (b) = Father _____

Ans: 3

Difficulty: 3

66. Child of Mother (c) = Father _____

Ans: 1 or 4

Difficulty: 3

67. Child of Mother (d) = Father _____

Ans: 2 or 5

Difficulty: 3

68. Child of Mother (e) = Father _____

Ans: 1

Difficulty: 3

100 *Nicotiana* plants were tested for self and cross sterility and were found to fall into groups which had the following relationships with respect to offspring:

Male Parent

	A	B	C	D
A	None	$\frac{1}{4}$ H $\frac{1}{4}$ E $\frac{1}{4}$ F $\frac{1}{4}$ G	$\frac{1}{4}$ H $\frac{1}{4}$ M $\frac{1}{4}$ F $\frac{1}{4}$ D	$\frac{1}{2}$ D $\frac{1}{2}$ M
B	$\frac{1}{4}$ H $\frac{1}{4}$ E $\frac{1}{4}$ F $\frac{1}{4}$ G	None	$\frac{1}{2}$ R $\frac{1}{2}$ C	$\frac{1}{4}$ H $\frac{1}{4}$ C $\frac{1}{4}$ R $\frac{1}{4}$ G
C	$\frac{1}{4}$ H $\frac{1}{4}$ D $\frac{1}{4}$ M $\frac{1}{4}$ F	$\frac{1}{2}$ R $\frac{1}{2}$ B	None	$\frac{1}{2}$ D $\frac{1}{2}$ F
D	$\frac{1}{2}$ A $\frac{1}{2}$ M	$\frac{1}{4}$ F $\frac{1}{4}$ C $\frac{1}{4}$ R $\frac{1}{4}$ G	$\frac{1}{2}$ H $\frac{1}{2}$ C	None

Identify the allelic constitutions of EACH:

69. A _____

Ans: S^1S^2

Difficulty: 4

70. B _____

Ans: S^3S^4

Difficulty: 4

71. C _____

Ans: S^3S^5

Difficulty: 4

72. D _____

Ans: S^2S^5

Difficulty: 4

73. E _____

Ans: S^1S^4

Difficulty: 4

74. F _____

Ans: S^2S^3

Difficulty: 4

75. G _____

Ans: S^2S^4

Difficulty: 4

76. H _____

Ans: S^1S^3

Difficulty: 4

77. M _____

Ans: S^1S^5

Difficulty: 4

78. R _____

Ans: S⁴S⁵

Difficulty: 4

These data represent the occurrence of the same gene in different mutants. It is sometimes referred to as complementation mapping. In the table, “O” indicates that both organisms are mutant for the same gene. The “+” indicates the mutants carry the mutation in different genes. The boxes on top of the lines indicate the mutants A through F. If the lines overlap then the two mutations are in the same gene. Use the data in the table to determine which boxes contain the appropriate mutation. Place the alphabet in the appropriate boxes on each line to show complementation (different genes) & non-complementation (same gene). Two answers have been provided: Mutants “A” and “F” are mutant in the same gene. Mutants “A” and “F” have a “O” in the table, therefore their lines should overlap.

	A	B	C	D	E	F
A	O	+	+	O	O	O
B	+	O	O	O	O	+
C	+	O	O	+	+	+
D	O	O	+	O	O	+
E	O	O	+	O	O	+
F	O	+	+	+	+	O

F

A

79.

Ans: B

Difficulty: 3

80.

Ans: D or E

Difficulty:

81.

Ans: C

Difficulty:

82.

Ans: D or E

Difficulty:

Genes “A” and “B” are required for color. If “A” or “B” are absent (“a” or “b”) the result is colorless. Give the genotypes and phenotypes for each F₁ and F₂ below.

	Parent	F1		F2	
		Genotype	Phenotype	Genotype	Phenotype
83.	AAbb × aabb				
84.	aaBB × aabb				
85.	AAbb × aaBB				

83. Aabb × aabb

Ans: F₁=Aabb/Colorless; F₂= 1AAbb:2Aabb:1aabb/All colorless

Difficulty: 2

84. aaBB × aaBB

Ans: F₁=aaBb/Colorless; F₂= 1aaBB:2aaBb:1aabb/All colorless

Difficulty: 2

85. AAbb × aabb

Ans: 9 Color; 7 Colorless

<u>Genotype</u>	<u>Phenotype</u>
F ₁ =AaBb	Color
F ₂ =1AABB	Color
2AABb	Color
2AaBB	Color
4AaBb	Color
1aaBB	Colorless
1AAbb	Colorless
2aaBb	Colorless
2Aabb	Colorless
1aabb	Colorless

Difficulty: 3

Coat color in a certain species of rabbit is governed by multiple alleles. The hierarchy with reference to dominance for these alleles are as follows: Colored (c⁺), Chinchilla, (c^{ch}), Himalayan (c^h) and Albino (c). Give the phenotypes and ratios from the following crosses.

86. c⁺ c × c^h c^h

Ans: 1 normal; 1 himalayan

Difficulty: 1

87. c⁺ c⁺ × c^h c^h

Ans: All normal

Difficulty: 1

88. $c^+ c \times c^h c$
 Ans: 2 normal; 1 himalayan; 1 albino
 Difficulty: 1
89. $c c \times c^h c^h$
 Ans: 1 himalayan; 1 chinchilla
 Difficulty: 1
90. $c^+ c^h \times c^h c^h$
 Ans: 2 normal; 2 himalayan
 Difficulty: 1
91. $c^+ c^{ch} \times c^h c^{ch}$
 Ans: 2 normal; 1 himalayan; 1 chinchilla
 Difficulty: 1
92. $c c \times c+c^h$
 Ans: 2 normal; 2 chinchilla
 Difficulty: 1
93. In a certain breed of plants, dark green is determined by the dominant gene "G" and light green is determined by the recessive gene "g". The heterozygote shows 75% penetrance and will look phenotypically light green. If the parental cross is $GG \times gg$; what phenotypes would be observed in a population of 400 F_2 plants?
 Ans: 250 Dark Green; 150 light green
 Difficulty: 4
94. A man with blood type "A," whose father was blood type "O," married a woman of blood type "B" whose mother was blood type "O." What are the possible blood types of their offspring?
 Ans: Blood Type A, B, AB and O are possible
 Difficulty: 1
95. What phenotypes and genotypes would you expect from the following cross.
 $I^B i rh^+ rh^- \times I^A i rh^+ rh^-$
 Ans: $I^B I^A rh^+ rh^-$ AB positive
 $I^B I^A rh^+ rh^+$ AB negative
 $ii rh^+ rh^-$ O positive
 $ii rh^+ rh^+$ O negative
 Difficulty: 2

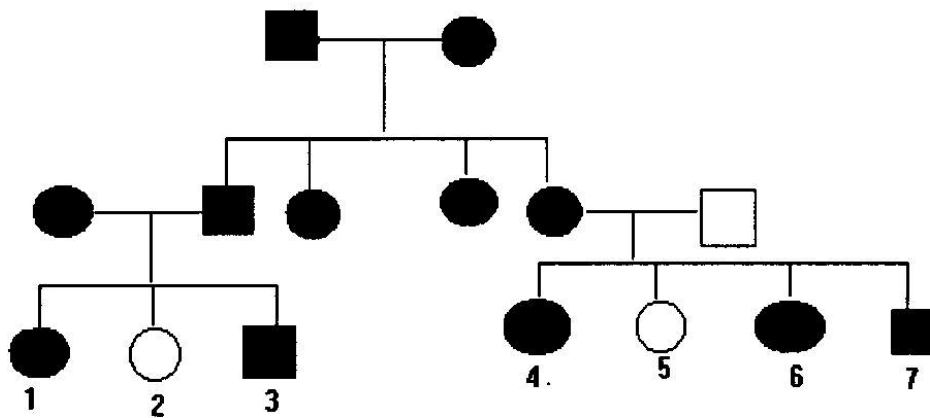
96. Horses can be cremello (light cream color), chestnut (a brownish color) or palomino (golden color with white in the horse's tail and mane). Of these phenotypes, only palominos never breed true. From the results shown below, determine the mode of inheritance by assigning gene symbols and indicating which genotypes yield which phenotypes.

Parents	F ₁ progeny	Genotypes of parents?
(A) Cremello × Palomino	½ cremello & ½ palomino	(A)
(B) Chestnut × Palomino	½ chestnut & ½ palomino	(B)
(C) Palomino × Palomino	¼ chestnut, ¼ cremello ½ palomino	(C)

Ans: A) PP × Pp / B) pp × Pp / C) Pp × Pp

Difficulty: 2

Below is a pedigree for a human trait. Shaded symbols are for individuals showing the trait. (A) Identify the mode of inheritance of the trait Dominant or Recessive. (B) Apply the laws of probability and calculate the probability the offspring of the cousin marriage will have the trait.



97. mode of inheritance _____

Ans: Dominant

Difficulty: 1

98. probability of 2 × 7 _____

Ans: 1/2

Difficulty: 2

99. probability 3 × 5 _____

Ans: 2/3

Difficulty: 3

100. probability 1×7 _____
Ans: 8/9 will have the trait
Difficulty: 4

Experimental Design and Interpretation of Data

101. You have obtained an interesting flower for your garden from your neighbor. The neighbor has given you two pure lines of the plant, one with red flowers and one with yellow flowers. You decide to cross them and find that you obtain all orange flowers. The curious molecular geneticist in you decides to test two independent hypotheses: Hypothesis 1: Incomplete Dominance; Hypothesis 2: Recessive Epistasis. The first step in your test is to self the F_1 orange plants, which you complete only to find that the results do not statistically distinguish the two hypotheses. a) What ratio of yellow, orange and red would you expect in the F_2 population for each hypothesis and b) what crosses would you complete next to definitively test your two hypotheses?

Ans: a) The expected phenotypic ratio for recessive epistasis is 9:3:4 and 1:2:1 for incomplete dominance. b) Cross the yellow F_2 flowers with true breeding red flowers. If the hypothesis for incomplete dominance is correct, the yellow color will be determined by a single gene and all F_2 yellow flowers will be homozygous recessive and give rise to only orange flowers in the F_3 population [$aa \times AA = Aa$]. However, if the hypothesis for recessive epistasis is correct, a cross of F_2 yellow and true breeding red flowers will give rise to some red and some orange flowers [$Yyrr \times yyRR =$ either $yyRr$ or $YyRr$].

Difficulty: 4

102. Your nephew has asked for help with his science project that focuses on coat color determination in mice. He has obtained several mice of different colors from the lab of a generous researcher at the local university. The two of you set out to breed the mice together in a variety of crosses to determine how coat color is inherited. Your curious and intelligent nephew notices that the mice have a variety of different tail lengths that make them easier or more difficult to pick up by the tail. He asks you about this oddity and in your search for answers you call the scientist from whom he received the mice only to find out that he has never paid much attention to this trait before but that the mice are inbred strains maintained by a national animal provider. When you call the company they tell you that all the members of each strain look the same but that the differences between the strains have always existed. Based on this information, you now know how the different tail lengths occur. What do you tell your nephew?

Ans: Tail length is genetically determined by a modifier gene (as opposed to a tail versus no tail) for which there are several alleles because the inbred mice always differ between strains.

Difficulty: 4

103. A friend of yours is a volunteer in the Peace Corps working in an area of the world that is extremely hot. Your friend has captured some interesting crème colored mice and adopted them as pets. When your friend returns to his home town in the northern Midwest in the U.S., he is surprised to find that his crème colored pet mice are now giving birth to babies that have dark brown ears and legs. Your friend knows that you are a genetics student and asks for your help to explain this phenomenon. Given that your friend is certain that no other mouse got into the cage to breed, what do you tell your friend about the genetics underlying his mouse results?

Ans: The mouse coat color is affected by temperature. The genetic term that describes this phenomenon is phenocopy, a change in phenotype that mimics a mutation but arises from an environmental effect.

Difficulty: 3

104. You are a judge in a civil trial where a young man is attempting to prove that he is the illegitimate child of a very wealthy man who has recently died. He wishes to be included in the distribution of his wealth. After considering all the testimony about how this person was conceived the key evidence seems to come down to two main facts. The wealthy man and the mother of the young man are both deaf but the young man is not. Therefore the lawyer of the family suggests that the wealthy man is not the father. The mother, wealthy man and young man all have O, MM, Rh- blood type at the phenotypic level but a genotyping screen indicates that the wealthy man is actually $I^A I^A hh$ blood type. How do you interpret the evidence presented and how does it influence your decision in this legal case?

Ans: The fact that the young man can hear is not evidence against his being the son of the wealthy man. Two deaf individuals can, via complementation, give rise to hearing offspring if the mutation they carry is on different genes (Hearing is a polygenic trait.). The blood type evidence is definitive in favor of the wealthy man not being the father of the young man. Although both putative parents and the son in question have O blood type, the wealthy man is genetically type A and phenotypically type O due to recessive homozygosity of the h allele which leads to Bombay phenotype; the protein to which the A sugar attaches is missing therefore making the wealthy man phenotypically type O. Any son of his would be highly likely to have A-antigen as the h allele is very rare in humans making homozygous recessive offspring extremely unlikely except in consanguineous matings.

Difficulty: 4

105. A science teacher is attempting to convince her class that alcoholism, which has long been known to be a disease of polygenic inheritance, really is partially genetically determined. You are asked to assist in the design of an experiment that will help show eighth graders genetic transmission of differences in alcohol drinking. You have been given outbred rats as your experimental model. Set up a quantitative experiment that would test the hypothesis that alcoholism, as determined by amount of alcohol drunk, is a quantitative trait.

Ans: Set up a selective breeding experiment. Measure alcohol consumption per day for all rats. Breed the high drinking male rats with the high drinking females and the low drinking males with low drinking females. Test the offspring and do the same in subsequent generations. If the rats bred for high drinking continue to increase their drinking levels from generation to generation and the low drinkers decrease their drinking levels in the same, this is evidence that alcohol drinking is genetically determined. Your data will also show that the individual rats differ in consumption and when plotted together show a continuous distribution indicating a quantitative trait (more than one gene interactions, and interactions with the environment contribute to the alcohol drinking trait).

Difficulty: 4